Code No. 8414

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (AI&ML) IV-Semester (MAIN) Examinations Jun/Jul-2026

ARTIFICIAL INTELLIGENCE

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

			Marks	BT	CO
1.	a)	Define Artificial Intelligence and mention any two applications of AI.	2	1	3
	b)	Differentiate between uninformed search and informed search strategies.	2	4	1
	c)	What is predicate logic? How is it different from propositional logic?	2	2	1
	d)	What is a Bayesian Belief Network?	2	2	2
	e)	What is distributed representation in connectionist models?	2	1	2
	f)	How do we handle uncertainty in AI?	2	1	1
2.	a)	Describe the structure of intelligent agents and explain different types of agents used in AI systems.	6	2	2
	b)	Discuss the nature of environments in AI and explain how performance measures influence rational agent behavior.	6	4	1
3.	a)	Discuss adversarial search in games and explain the minimax algorithm with alpha beta pruning and iterative deepening.	6	5	1
	b)	Explain informed search strategies and discuss the role of heuristic functions in problem solving.	6	2	1
4.	a)	Explain different approaches to knowledge representation in artificial intelligence.	6	2	1
	b)	Discuss logic programming and explain its importance in AI.	6	3	1

(P.T.O.)

5.	a)	Explain the architecture of an expert system and discuss the phases involved in building expert systems.	6	2	2
	b)	Explain Neuro-fuzzy systems and discuss their advantages and applications in artificial intelligence.	6	5	2
6.	a)	Explain recurrent neural networks and describe their applications in artificial intelligence.	6	2	2
	b)	Compare connectionist AI and symbolic AI with their advantages and limitations.	6	4	3
7.	a)	Using axiomatic approach deduce A → C from {A → B, B →C}	6	3	1
	b)	Explain about the characteristics of problem and types of problems.	6	2	1

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